

A decorative graphic on the left side of the slide, consisting of a network of thin, light blue lines and small circles, resembling a circuit board or a neural network diagram. The lines are vertical and horizontal, with some diagonal connections, and the circles are small and light blue.

RESEARCH DESIGN

-NILAM SANGROULA



TAKEAWAYS

- MEANING AND IMPORTANCE OF RESEARCH DESIGN
 - CLASSIFICATION OF RESEARCH DESIGN
 - THE RESEARCH PROCESS
- ERRORS AFFECTING RESEARCH DESIGN
 - MEASUREMENT AND SCALING
 - RELIABILITY AND VALIDITY

MEANING OF RESEARCH DESIGN

- Jahoda, Deutch & Cook “A research design is the arrangement of conditions for the collection and analysis of data in a manner that aims to combine relevance to the research purpose with economy and procedure”
- The decisions like what, where, when, how, etc., in regard to a research study, creates a research design.
- In fact, the research design is the conceptual structure within which a research is conducted. Moreover, it comprises the outline for the collection, measurement and analysis of data.
- Hence, the design carries a blueprint of what the researcher will do.

THE RESEARCH DESIGN HAPPEN TO BE IN RESPECT OF

- What is the research?
- Where and why will the research be conducted?
- What data is required for the research?
- Where can be the data found?
- What will be the time period of the research?
- What will be the sample design?
- What methods will be used for data collection?
- How will be the data analysed?

NEED FOR RESEARCH DESIGN

- Research design is needed because it gives the smooth going of the various research operations, so making research as efficient as possible we can get maximal information with minimal expenditure of effort, time and money.
- Just as for better, economical and attractive construction of a house, we need a blueprint (or what is commonly called the map of the house) well thought out and prepared by an expert architect, similarly we need a research design or a plan in advance of data collection and analysis for our research project.

TYPES OF RESEARCH DESIGN

- **Quantitative Research Design:**

- Quantitative research is referred to as the process of collecting as well as analyzing numerical data. It is generally used to find patterns, averages, predictions, as well as cause-effect relationships between the variables being studied. It is also used to generalize the results of a particular study to the population. Quantitative research provides actionable insights which are essential for the growth of the company.

- **Qualitative Research Design:**

- Qualitative research can be defined as a method used for market research which aims at obtaining data through open-ended questions and conversations with the intended consumers. This method aims at establishing not only “what” people think but “how”

SUB-TYPES OF RESEARCH DESIGN

- Descriptive research: It is defined as a research method that describes the characteristics of the population or phenomenon studied. This methodology focuses more on the “what” of the research subject than the “why” of the research subject.
- In other words, it “describes” the subject of the research, without covering “why” it happens.



•Eg

- A fashion brand that wants to understand the fashion purchasing trends among Birtamode buyers will conduct a demographic survey of this region, gather population data and then conduct descriptive research on this demographic segment. The study will then uncover details on “what is the purchasing pattern of Birtamode buyers,” but not cover any investigative information about “why” the patterns exists.
- 
- 
- 

CHARACTERISTICS OF DESCRIPTIVE RESEARCH

- **Quantitative research:** Descriptive research is a quantitative research method that attempts to collect quantifiable information for statistical analysis of the population sample. It is a popular market research tool that allows us to collect and describe the demographic segment's nature.
- **Uncontrolled variables:** In descriptive research, none of the variables are influenced in any way. This uses observational methods to conduct the research. Hence, the nature of the variables or their behavior is not in the hands of the researcher.
- **Cross-sectional studies:** Descriptive research is generally a cross-sectional study involves looking at data from a population at one specific point in time.
- **The basis for further research:** Researchers further research the data collected and analyzed from descriptive research using different research techniques. The data can also help point towards the types of research methods used for the subsequent research.

APPLICATIONS OF DESCRIPTIVE RESEARCH WITH EXAMPLES

- **Define respondent characteristics:** The aim of using close-ended questions is to draw concrete conclusions about the respondents. This could be the need to derive patterns, traits, and behaviors of the respondents. It could also be to understand from a respondent, their attitude, or opinion about the phenomenon. For example, understanding from adults the hours per week they spend on browsing the internet. All this information helps the organization researching to make informed business decisions.
- **Measure data trends:** Researchers measure data trends over time with a descriptive research design's statistical capabilities. Consider if an apparel company researches different demographics like age groups from 24-35 and 36-45 on a new range launch of autumn wear. If one of those groups doesn't take too well to the new launch, it provides insight into what clothes are like and what is not. The brand drops the apparel that customers don't like.

CONT.

- **Conduct comparisons:** Organizations also use a descriptive research design to understand how different groups respond to a specific product or service. For example, a fashion brand creates a survey asking general questions that measure the brand's image. The same study also asks demographic questions like age, income, gender, geographical location, etc.
- **Conduct research at different times:** The analysis can be conducted at different periods to ascertain any similarities or differences. This also allows any number of variables to be evaluated (Longitudinal Research)

EXPLORATORY RESEARCH DESIGN:

- Exploratory Study is undertaken when not much is known about the situation at hand, or no or very little information is available on how similar problems or research issues have been solved in the past.
- For such a research, a researcher starts with a general idea and uses this research as a medium to identify issues, that can be the focus for future research. Such research is usually carried out when the problem is at a preliminary stage.
- Eg.
 - The manager of a multinational corporation is curious to know if the work ethics values of employees working in AL Hassan Industrial City would be different from those of Americans. That city is a small city, and no information about the ethic values of its workers. Also, the work ethical values mean be different to people in different

EXPLANATORY RESEARCH DESIGN

- Researchers often aim to explain why particular phenomena work in the way that they do. Research that answers “why” questions is referred to as **explanatory** research. In this case, the researcher will try to identify the causes and effects of whatever phenomenon they are studying.
 - A study of college students’ addictions to their electronic gadgets might aim to understand why students become addicted. Does the addiction have anything to do with their family histories, extracurricular hobbies and activities, or with whom they spend their time? An explanatory study could answer these kinds of questions.

DIAGNOSTIC RESEARCH DESIGN:

- In a diagnostic research design, the researcher will try to evaluate the cause of a specific problem or phenomenon. This research design is used to understand more in detail about the factors that are creating problems in the company. Diagnostic research design includes three steps:
 - Inception of the issue – When did the issue arise? In what situations is the issue more evident?
 - Diagnosis of the issue – What is the underlying cause of the issue? What is influencing the issue to worsen?
 - Solution for the issue – What is working in curing the issue? Under what situations does the problem seem to become less evident?

CROSS-SECTIONAL DESIGN

- A cross-sectional design is used for research that collects data on relevant variables one time only from a variety of people, subjects, or phenomena.
- A cross-sectional design provides a snapshot of the variables included in the study, at one particular point in time.
- Cross-sectional designs generally use survey techniques to gather data, for example, the Nepal Census, 2021.

LONGITUDINAL DESIGN

- A longitudinal design collects data over long periods of time.
- Measurements are taken on each variable over two or more distinct time periods.
- This allows the researcher to measure change in variables over time.

TIME SERIES DESIGN

- A Time Series Design collects data on the same variable at regular intervals in the form of aggregate measures of a population.
- Time series designs are useful for:
 - establishing a baseline measure
 - describing changes over time
 - keeping track of trends

POTENTIAL ERRORS AFFECTING RESEARCH DESIGN

- **Population Specification**

- Population specification errors occur when the researcher does not understand who they should survey.

- Eg.

- Foods manufacturers often conduct surveys of housewives, because they think they decide what is to be purchased and also do the actual purchasing. In this situation there often is population specification error. The husband may purchase a significant share of the foods, and have significant direct and indirect influence over what is bought. For this reason, excluding husbands from samples may yield results targeted to the wrong audience.

SAMPLING ERRORS

- Sampling error arises from the fact that a sample, rather than the entire population, is used to make inferences about the population. It is the error caused by observing a sample instead of the whole population.
- **Causes of Sampling Error:**
 - **Sample Size:** Smaller samples tend to have higher sampling errors.
 - **Sampling Method:** Non-random sampling methods can lead to biased samples.
- **Examples:**
 - If a political poll surveys 1,000 voters out of millions, the results will have a sampling error because it may not perfectly represent the whole voter population.
 - A clinical trial with 50 participants may yield different results compared to a trial with 5,000

NON-SAMPLING ERRORS

- Non-sampling errors are errors that are not related to the act of sampling itself. These errors can occur in the process of data collection, processing, or analysis, and can affect the validity of the results even if the entire population is surveyed.

TYPES OF NON-SAMPLING ERRORS:

- **Measurement Error:** Inaccuracies in measuring the variables of interest.
 - Example: A poorly calibrated scale gives incorrect weight measurements.
- **Data Processing Error:** Mistakes during data entry, coding, or analysis.
 - Example: Typographical errors during data entry.
- **Nonresponse Error:** When a significant portion of the sample does not respond.
 - Example: If many people do not answer a survey, the results may be biased.
- **Response Error:** When respondents provide inaccurate answers.
 - Example: People may lie or misremember facts when answering sensitive survey questions.
- **Coverage Error:** When some members of the population are not included in the sample frame.

MANAGING ERRORS IN RESEARCH DESIGN

- To minimize these errors, researchers can adopt various strategies:
- **Increase Sample Size:** Larger samples can reduce sampling error.
- **Random Sampling:** Ensures that every member of the population has an equal chance of being included.
- **Pre-testing Instruments:** Helps in identifying and correcting measurement errors.
- **Training Data Collectors:** Ensures data is collected consistently and accurately.
- **Follow-Up on Non-respondents:** Helps to reduce nonresponse error.
- **Data Validation:** Double-checking and validating data entries can

MEASUREMENTS AND SCALING

- Assigning numbers or other symbols to any product, event, issues or characteristics as per the certain pre-specified rule is known as measurement.
- Number is a symbol that can be provided in different ways like 1,2,3.
- Number should be explained to get the information.
- These assignments of the number to the product, event, issues or characteristics should reflect the character.
- It is very easy to measure the tangible products like length, breadth, height and weight etc.
- It is very difficult to measure the intangible things and hence it is essential to provide number or symbol to measure the qualitative variables.
- Thus: measurement permits statistical analysis and facilitates the communication

DEFINITION OF MEASUREMENT

- Goode and Hatt, “ Measurement is the method of turning the series of qualitative facts into a quantitative series.”
- S. Steven, “ Measurement is the equipment of providing numbers to objects or events according to rule.”
- Cooper and Schindler, “ Measurement in research consists of assigning numbers to empirical events, objects or properties or activities in compliance with a set of rules.”
- Thus, measurement is that sort of tool which provides series of numbers to the series of characteristics of event, product or things.

SCALES OF MEASUREMENT

- In business research/social science, the below are the basic types of scales
 - Nominal Scale
 - Ordinal (Rank) Scale
 - Interval Scale
 - Ratio Scale
- The data collected for the research study will fit into one of these scales.
- It is, therefore, important to understand the different levels of data to do a proper analysis.

NOMINAL SCALE

- A nominal scale is one that allows the researcher to assign subjects to certain categories of groups.
- Nominal stands for “Name of category”
- Numbers are used simply as labels for groups or class
- It is used for qualitative rather than quantitative data
- Nominal scale is the least powerful level of measurement
- Nominal scales provide convenient ways of keeping track of people, objects and events
- One cannot do much with the numbers involved



EG.

- Numbers on the t-shirts of team players
 - One cannot usually average the number on the back of a group of players and come of with the meaningful value
 - The counting of members in each group is the only possible arithmetic operation when a nominal scale is employed.
- 
- 
- 

FOR INSTANCE, A CUSTOMER SURVEY ASKING “WHICH BRAND OF SMARTPHONES DO YOU PREFER?” OPTIONS : “APPLE”- 1 , “SAMSUNG”-2, “ONEPLUS”-3.

- In this survey question, only the names of the brands are significant for the researcher conducting consumer research. There is no need for any specific order for these brands. However, while capturing nominal data, researchers conduct analysis based on the associated labels.
- In the above example, when a survey respondent selects Apple as their preferred brand, the data entered and associated will be “1”. This helped in quantifying and answering the final question – How many respondents selected Apple, how many selected Samsung, and how many went for OnePlus – and which one is the highest.
- This is the fundamental of quantitative research, and nominal scale is the most

ORDINAL SCALE

- Ordinal scale is defined as a variable measurement scale used to simply depict the order of variables and not the difference between each of the variables.
- These scales are generally used to depict non-mathematical ideas such as frequency, satisfaction, happiness, a degree of pain, etc. It is quite straightforward to remember the implementation of this scale as 'Ordinal' sounds similar to 'Order', which is exactly the purpose of this scale.
- This scale not only assigns values to the variables but also measures the rank or order of the variables, such as:



E.G.

- What is the highest degree or level of school you have completed?
 - Less than a high school diploma
 - High School degree or equivalent
 - Some college, no degree
 - Associate degree
 - Bachelor's degree
 - Master's degree
- 
- 
- 

INTERVAL SCALE

- Interval data refers not only to classification and ordering the measurements, but it also specifies that the distances between each value on the scale are **equal**. The distance between values is meaningful.
- To put it another way, the differences between points on the scale are equivalent. That is why it is called interval data. It is measured on interval scales. The Interval scale is a numeric scale that represents not only the order but also the equal distances between the values of the objects.
- When you ask whether it's hot, warm, or cold outside, you're using an ordinal scale. When you ask what the temperature is, you're using an interval scale.

INTERVAL SCALE

- Interval data refers not only to classification and ordering the measurements, but it also specifies that the distances between each value on the scale are **equal**. The distance between values is meaningful.
- To put it another way, the differences between points on the scale are equivalent. That is why it is called interval data. It is measured on interval scales. The Interval scale is a numeric scale that represents not only the order but also the equal distances between the values of the objects.
- When you ask whether it's hot, warm, or cold outside, you're using an ordinal scale. When you ask what the temperature is, you're using an interval scale.

- Any survey question where the respondent must enter a temperature value, time, or date, their likelihoods technically includes an interval scale, since degrees, hours, and days are all interval measurements. This means you can easily insert an interval scale into any multiple choice questions or rating scale questions. You just need to ask respondents to rate something using a numerical scale.
- It is used in various sectors like in education, medicine, engineering, etc. Some of these uses include calculating a student's CGPA, measuring a patient's temperature , etc.



EG.

- How satisfied are you with our service on a scale from 1 to 10, where 1 means 'Not at all satisfied' and 10 means 'Extremely satisfied'?"
 - How often do you use our product on a scale from 1 to 10, where 1 means 'Never' and 10 means 'All the time'?"
 - How likely are you to recommend our product to a friend on a scale from 0 to 10, where 0 means 'Not at all likely' and 10 means 'Extremely likely'?"
- 
- 
- 

RATIO SCALE

- The ratio scale is the 4th level of measurement scale. A ratio scale is a quantitative scale where there is a true zero and equal intervals between neighboring points. It allows researchers to compare the differences or intervals. The ratio scale has a unique feature. It possesses the character of the origin or zero points.
- **Characteristics of Ratio Scale:**
 - Ratio scale has a feature of absolute zero
 - It doesn't have negative numbers, because of its zero-point feature
 - It affords unique opportunities for statistical analysis. The variables can be orderly added, subtracted, multiplied, divided. Mean, median, and mode can be calculated using the ratio scale.
 - Ratio scale has unique and useful properties. One such feature is that it allows



E.G.

•What is your weight in Kgs?

- Less than 55 kgs
 - 55 – 75 kgs
 - 76 – 85 kgs
 - 86 – 95 kgs
 - More than 95 kgs
- 

RATING SCALES

- The following rating scales are often used in research
 - Dichotomous scale
 - Semantic differential scale
 - Category scale
 - Numerical scale
 - Itemized rating scale
 - Likert scale
 - Fixed or constant sum rating scale
 - Staple scale
 - Graphic rating scale

DICHOTOMOUS SCALE

- A dichotomous scale is a type of survey response scale that provides two options, which lie at opposite ends. On a dichotomous scale, the survey respondent can not give a neutral answer because it is a case of either one or the other.
- E.g.
 - Did you enjoy using our product?
 - Yes
 - No
 - Was this article helpful to you?
 - Yes
 - No

SEMANTIC DIFFERENTIAL SCALE

- Ratings are basically “oscillating” to the extent that they display two adverse characteristics of a subject under study (for example, low to high scale or good to bad scale). These survey answering options are grammatically on opposite adjectives at each end.

- For example, love / hate, satisfied / unsatisfied and likely to return / unlikely to return with intermediate options in between.

- *It seems like you purchased our new robotic vacuum cleaner a month ago. How would you rate it on the following criteria?*

• Inconvenient Appliance - - - - - Convenient Appliance

• Expensive Purchase. - - - - - Inexpensive Purchase

CATEGORY SCALE

- The category scale uses multiple items to elicit a single response. E.g.
- Where in Jhapa do you reside?
 - Birtamode
 - Surunga
 - Budhabare
 - Charpane

NUMERICAL SCALE

- The numerical Scale is similar to the semantic differential scale, with the difference that numbers on a five or seven point scale are provided, with bipolar adjectives at both ends.

- How pleased are you with your new teacher?

• Extremely pleased 7 6 5 4 3 2 1 Extremely
Displeased

LIKERT SCALE OF MEASUREMENT

- Likert scale of measurement is widely used scale in research
- A Likert-type scale assumes that the strength of experience is linear, i.e. on a continuum from strongly disagree to strongly agree, and makes the assumption that attitudes can be measured.
- Respondents may be offered a choice of five pre-coded responses with the neutral point being neither agree nor disagree.
- The Likert Scale is a five point scale which is used to allow the individual to express how much they agree or disagree with a particular statement.
 - I believe that ecological questions are the most important issues facing human beings today.

STAPEL SCALE

- Stapel Scale is a unipolar (one adjective) rating scale designed to measure the respondent's attitude towards the object or event. The scale is comprised of categories ranging from -5 to +5 or -3 to +3 without any neutral point (zero).
- The scale is usually constructed vertically with a single adjective in the middle of the range of values (-5 to +5 or -3 to +3). The respondent is asked to select the appropriate numerical response category that best describes the extent to which the adjective related to the object is accurate or inaccurate.

E.G.

- Please rate the quality of food, and crew member service of an airline on a scale ranging from -5 to +5:

+5	+5
+4 X	+4
+3	+3 X
+2	+2
+1	+1
High Food Quality	Poor Crew Service
-1	-1
-2	-2
-3	-3
-4	-4
-5	-5

RELIABILITY AND VALIDITY OF A SCALE

- **Reliability:**
- Reliability refers to the consistency and stability of a measurement instrument. In other words, it assesses whether the scale produces consistent results when administered multiple times.
- Two common approaches to assessing reliability are:
 - Internal consistency assesses how consistently the items within a scale measure the same construct. It ensures that the items are interrelated and contribute to a reliable measurement.
 - Commonly used measure: Cronbach's alpha (α).
 - Example: Suppose you create a 10-item anxiety scale. Administer it to a group of participants, and then calculate Cronbach's alpha. A high α (e.g., 0.80 or above)

• **Test-Retest Reliability:**

- Test-retest reliability evaluates the stability of a scale over time. It involves administering the same scale to the same group of participants on two separate occasions.
- Example: Imagine you develop a happiness scale. You administer it to a group of individuals today (Time 1) and then again after two weeks (Time 2). High correlation between the scores at Time 1 and Time 2 suggests good test-retest reliability.

VALIDITY

- Validity refers to the extent to which a scale measures what it claims to measure. It ensures that the scale captures the intended construct accurately.
- Different types of validity include:
- **Content Validity:**
 - Content validity ensures that a measurement scale adequately covers all relevant aspects of the construct it intends to measure.
 - Example: Imagine developing a depression assessment scale. To establish content validity, experts review the scale's items to ensure they comprehensively represent various symptoms of depression (e.g., sadness, loss of interest, sleep disturbances).



• **Criterion Validity:**

- Criterion validity assesses how well a scale correlates with an external criterion (e.g., an established gold standard or another valid measure).
- Example: Suppose you create a new intelligence test. To demonstrate criterion validity, you administer both your test and an existing widely accepted intelligence test to the same group of participants. High correlation between the two scores indicates good criterion validity.

• **Construct Validity:**

- Construct validity evaluates whether a scale measures the underlying construct it claims to measure.
 - Example: A self-esteem scale should correlate positively with related constructs (e.g., self-confidence, positive self-image) and negatively with unrelated constructs (e.g., anxiety). These relationships support construct validity.
- 

- **Convergent and Discriminant Validity:**

- Convergent validity examines whether a scale correlates with other measures of the same construct.
- Discriminant validity assesses whether the scale correlates less with unrelated constructs.
- Example: In a study on job satisfaction, you use a newly developed job satisfaction scale. High correlations with related constructs (e.g., work engagement) demonstrate convergent validity, while low correlations with unrelated constructs (e.g., physical health) show discriminant validity.

- **Face Validity:**

- Face validity is a subjective assessment of whether a scale appears to measure the intended construct.
- Example: When designing a survey about work-life balance, you ask potential users (employees) to review the items and provide feedback. Their perception of whether the items capture work-life balance contributes to face validity.

- **Predictive Validity:**

- Predictive validity determines whether a scale predicts future outcomes as expected.
- 
- 

ISSUES GOVERNING RESEARCH DESIGN

- **Research Questions and Objectives:**

- Clearly defining the research questions and objectives is crucial. This helps in determining the most appropriate research design and methods to use.
- Ensure that the questions are specific, measurable, achievable, relevant, and time-bound (SMART).

- **Research Design Types:**

- **Qualitative Research:** Focuses on understanding phenomena from a subjective perspective. Common methods include interviews, focus groups, and observations.
- **Quantitative Research:** Involves the collection and analysis of numerical data to identify patterns and test hypotheses. Methods include surveys, experiments, and secondary data analysis.
- **Mixed-Methods Research:** Combines qualitative and quantitative approaches to



• Sampling:

- **Population and Sample:** Define the target population and determine how the sample will be selected. Sampling methods can be probability (random sampling) or non-probability (convenience, purposive sampling).
 - **Sample Size:** Determining an appropriate sample size is critical for ensuring the validity and reliability of the study.
- 
- 



• **Ethical Considerations:**

- Obtain informed consent from participants.
 - Ensure confidentiality and anonymity.
 - Address any potential risks to participants.
 - Seek approval from relevant ethics committees or institutional review boards (IRBs).
- 
- 



• **Validity and Reliability:**

- **Validity:** Ensure that the research design accurately measures what it intends to measure. This includes internal validity (causality) and external validity (generalizability).
- **Reliability:** Ensure that the research methods produce consistent and repeatable results.

• **Data Collection Methods:**

- Choose appropriate data collection methods based on the research questions and design. This could include surveys, interviews, observations, or secondary data sources.
 - Consider the tools and instruments used for data collection, ensuring they are valid and reliable.
- 
- 



• **Data Analysis:**

- Plan for the appropriate data analysis techniques, whether statistical analysis for quantitative data or thematic analysis for qualitative data.
- Use software tools where necessary, such as SPSS for quantitative analysis or NVivo for qualitative analysis.

• **Bias and Limitations:**

- Identify and address potential sources of bias in the research design, data collection, and analysis.
 - Acknowledge the limitations of the study and how they might impact the findings.
- 
- 



• **Operationalization:**

- Clearly define how abstract concepts will be measured and observed in the study.
- Ensure that the operational definitions align with the theoretical framework.

• **Research Plan and Timeline:**

- Develop a detailed research plan and timeline to manage the different phases of the study effectively.
 - Include milestones and deadlines to ensure timely completion.
- 
- 



•Resource Management:

- Assess the resources required for the study, including funding, personnel, equipment, and time.
 - Ensure that the resources are available and managed efficiently.
 - By addressing these issues, researchers can design studies that are methodologically sound, ethically responsible, and capable of producing valid and reliable results.
- 
- 